SUMScore

Less repeated computation. Comparable retrieval quality.

Optimizing long-document retrieval with cached summaries in RAG systems

CHALLENGE 02

Deliver AI with Less Compute

Research foundation: Eltahir et al., Periscope (2026)

The problem: accuracy vs. processing cost

EMBEDDING-BASED RAG

Fast retrieval

But semantic similarity can miss reasoning-intensive relevance.

LONG-CONTEXT LLM

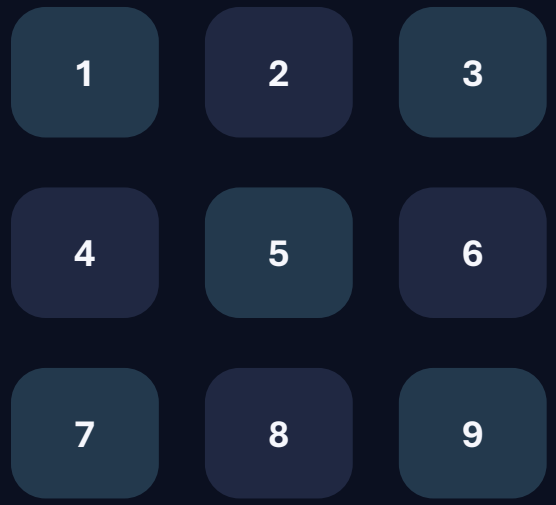
Deeper reasoning

But finite context windows and long-input inference can be expensive.

Target: reasoning-aware relevance at a lower repeated inference cost

Periscope solved the long-document read

KAUST research foundation • Eltahir et al. (2026)



K rows

local, nearby chunks

+

K columns

global, strided chunks

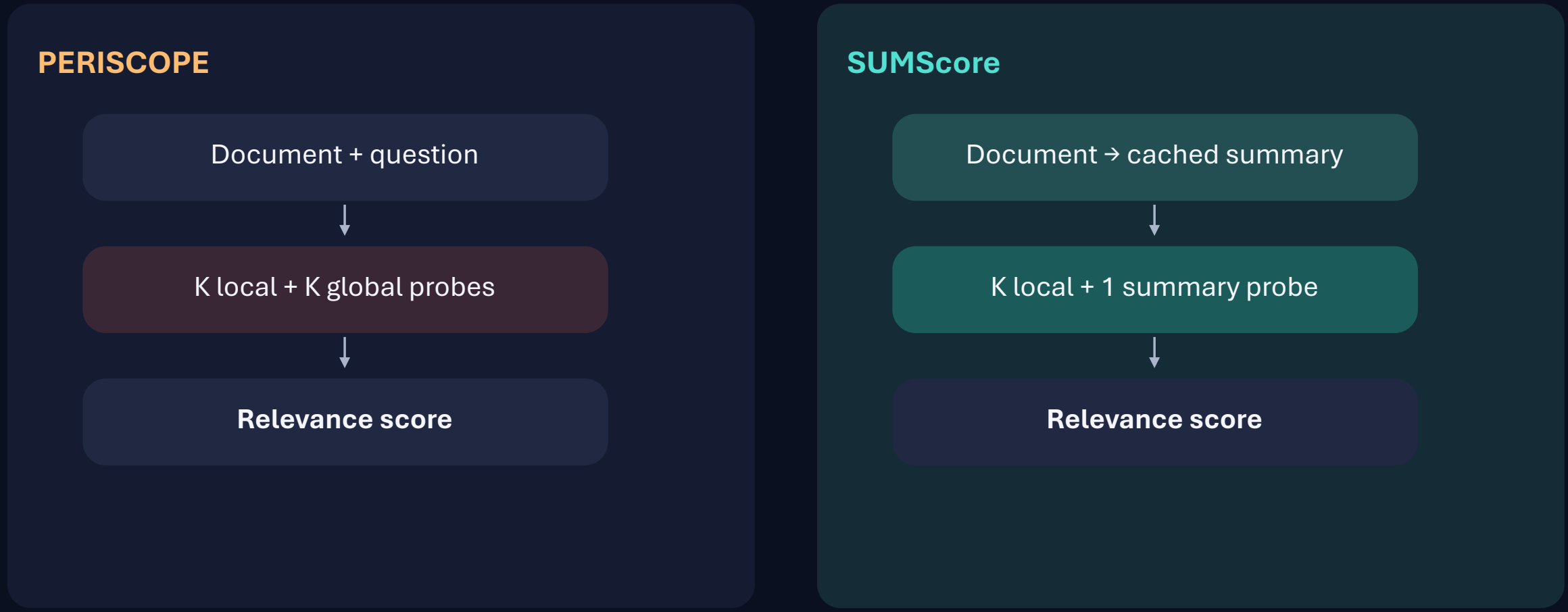
2K

probes / document / query

Opportunity: those global probes are recomputed for every new query.

Source: Periscope preprint, arXiv:2610.04047, §3

My contribution: reuse the global view



Reducing repeated AI inference in long-document retrieval through cached summaries

Document summaries are generated once; local probes and the cached summary are scored separately for each query.

Why this is an optimization solution

I optimize a measurable trade-off:

MINIMIZE

Online relevance calls + prompt tokens

SUBJECT TO

Comparable retrieval quality

Challenge: reduce unnecessary processing

Evidence: quality + cost measured together

Challenge 2 alignment: “cost per successful task” + “accuracy and latency floors”

Evaluation: reasoning-heavy retrieval

BRIGHT benchmark

Su et al. (2025) • reasoning-intensive search

200 questions

4 domains

100 candidate documents / query

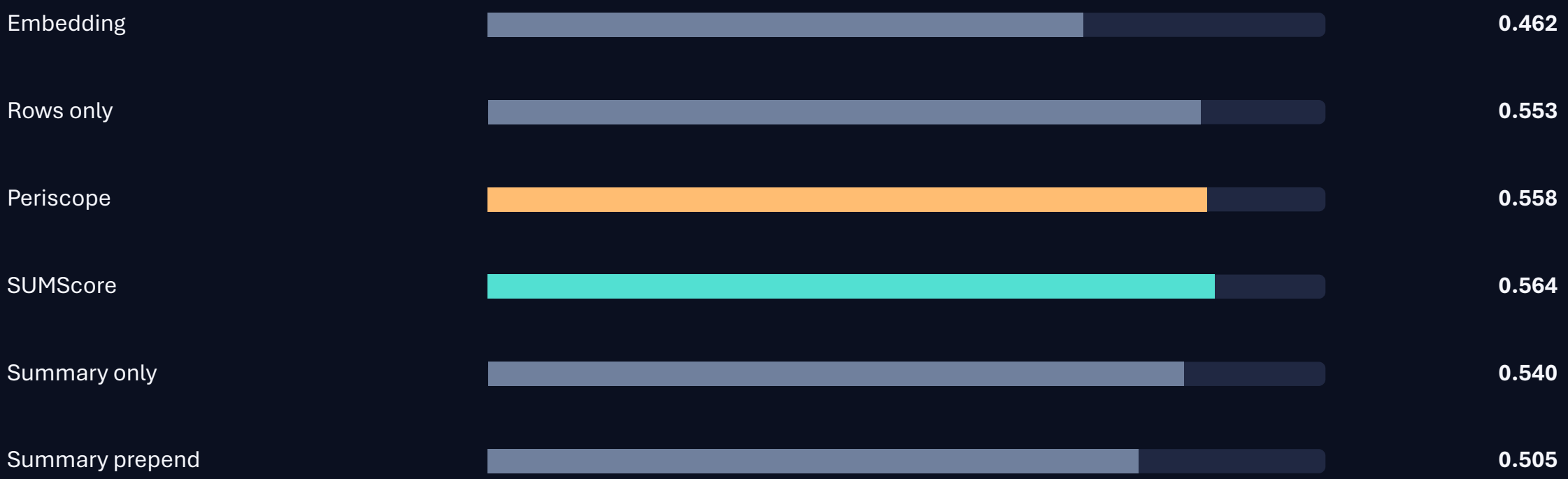
Six methods isolate the summary's contribution

Embedding · Rows only · Periscope · SUMScore (vote) · Summary prepend · Summary only

TECHNOLOGIES USED / IMPLEMENTATION
LLM: Qwen3-4B-Instruct-2507
Embedding model: Qwen3-Embedding-4B
Technologies: Python + vLLM
Chunk size: 100 tokens
Context window: 16K tokens

Same candidate pools across methods. This is a top-100 reranking experiment, not full-corpus retrieval.

Result 1 / Comparable retrieval quality

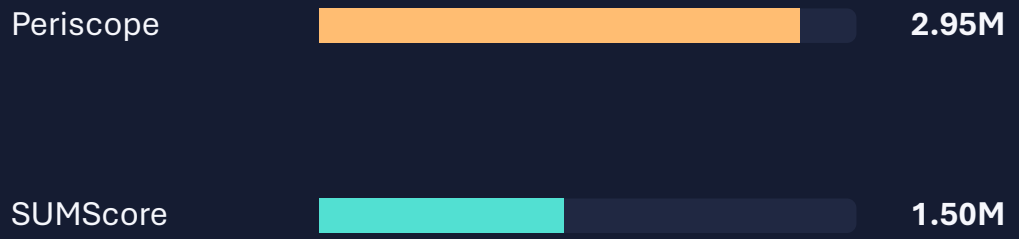


NDCG@10 ↑ | 200 BRIGHT questions; macro mean across four domains

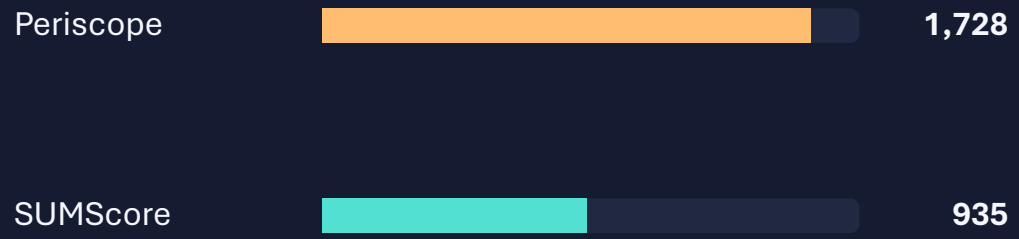
SUMScore vs. Periscope: +0.006; 95% paired CI [-0.023, 0.035] → difference not conclusive

Result 2 / Less repeated online processing

PROMPT TOKENS / QUERY



LLM CALLS / QUERY



49% fewer prompt tokens

46% fewer LLM calls

Online reranking metrics only. Offline summary creation is an additional one-time expense.

Value, limitations & next steps

WHAT THE TEST SUPPORTS

- ✓ Less repeated online inference
- ✓ Comparable NDCG@10
- ✓ Clear comparison with simpler alternatives

WHAT I WOULD TEST NEXT

Adaptive Retrieval
Automatically choose between lightweight retrieval and SUMScore based on query complexity.

Intelligent Cache Management
Reuse summaries across queries and refresh them only when documents change.

Project conclusion: **SUMScore demonstrates that reasoning-aware retrieval in RAG can become more efficient by reusing knowledge instead of repeatedly processing it.**

Prototype, code & references

Reyam Albalihi • SUMScore • HUMAIN Challenge 2

DEMO VIDEO	https://drive.google.com/file/d/1YSmH2L8ILhAzJfrJst6ZnAMZqFXnFJcb/view?usp=sharing
GITHUB	https://github.com/reyamm/SUMScope

Research: Eltahir et al. (2026), Periscope, arXiv:2610.04047

Dataset: BRIGHT (Su et al., 2025) • Testing and implementation: SUMScore notebook